3.0.41

Probability

CE 2008

Question

A person on a trip has a choice between private car and public transport. The probability of using a private car is 0.45. While using public transport, further choices available are bus and metro, out of which the probability of commuting by a bus is 0.55. The probability *(rounded up to two decimals)* of using a car, bus and metro, respectively would be:

a) 0.45, 0.30 and 0.25

b) 0.45, 0.25 and 0.30

c) 0.45, 0.55 and 0.00

d) 0.45, 0.35 and 0.20

Solution

Define the events:

$$A = \text{person uses private car} \tag{1}$$
$$A^c = \text{person uses public transport} \tag{2}$$
$$B = \text{person uses bus} \tag{3}$$
$$B^c = \text{person uses metro} \tag{4}$$

From the problem statement:

$$P(A) = 0.45 \tag{5}$$
$$P(A^c) = 1 - P(A) = 1 - 0.45 = 0.55 \tag{6}$$
$$P(B \mid A^c) = 0.55 \tag{7}$$
$$P(B^c \mid A^c) = 1 - P(B \mid A^c) = 1 - 0.55 = 0.45 \tag{8}$$

Solution

Using the multiplication rule of probability:

$$P(\mathrm{B}) = P(A^c) \cdot P(B \mid A^c) \tag{9}$$
$$= 0.55 \times 0.55 \tag{10}$$
$$= 0.3025 \approx 0.30 \tag{11}$$

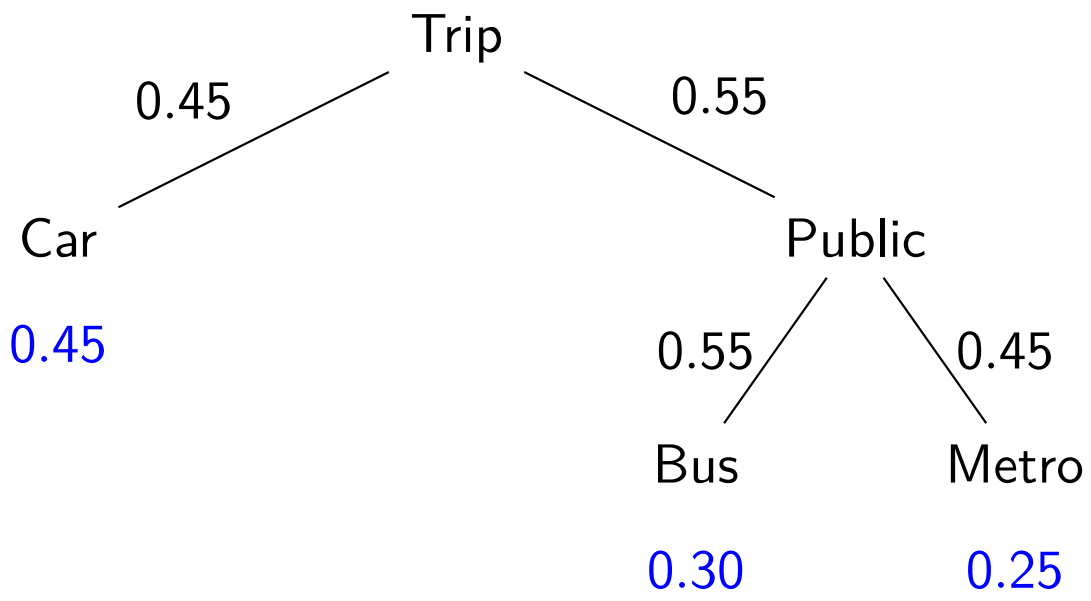
$$P(B^c) = P(A^c) \cdot P(B^c \mid A^c) \tag{12}$$
$$= 0.55 \times 0.45 \tag{13}$$
$$= 0.2475 \approx 0.25 \tag{14}$$

Solution

Verification — all probabilities must sum to 1:

$$P(A) + P(\mathrm{B}) + P(B^c) = 0.45 + 0.30 + 0.25 = 1.00 \quad \checkmark \qquad (15)$$

The tree diagram of the sample space:



Answer

Option (a): 0.45, 0.30 and 0.25

Plot

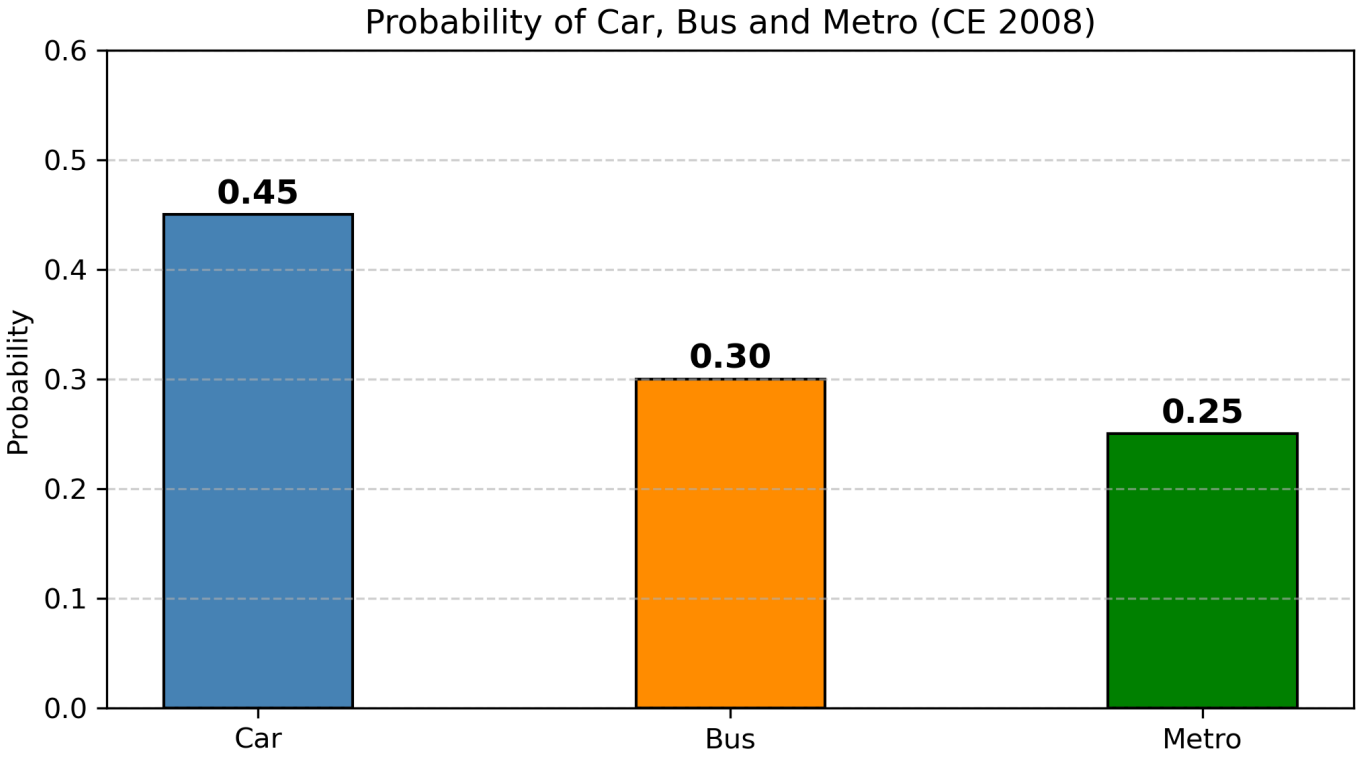


Figure: Probability tree diagram

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

P_A = 0.45
P_Ac = 1 - P_A

P_bus_given_pub = 0.55
P_metro_given_pub = 1 - P_bus_given_pub

P_bus = P_Ac * P_bus_given_pub
P_metro = P_Ac * P_metro_given_pub

labels = ['Car', 'Bus', 'Metro']
probs = [P_A, round(P_bus, 2), round(P_metro, 2)]
colors = ['steelblue', 'darkorange', 'green']
```

Python Code

```
plt.figure(figsize=(7, 4))
bars = plt.bar(labels, probs, color=colors, edgecolor='black',
    width=0.4)

for bar, val in zip(bars, probs):
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 0.01,
             f'{val:.2f}', ha='center', fontsize=12, fontweight='
                 bold')

plt.ylim(0, 0.6)
plt.ylabel('Probability')
plt.title('Probability of Car, Bus and Metro (CE 2008)')
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.savefig("image.png", dpi=300)
plt.show()
```